

From ODEs to the Chemical Master Equation to Gillespie Simulation

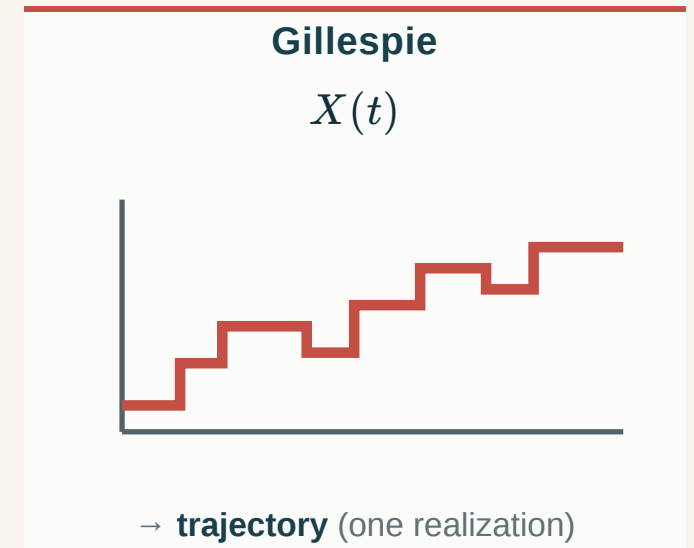
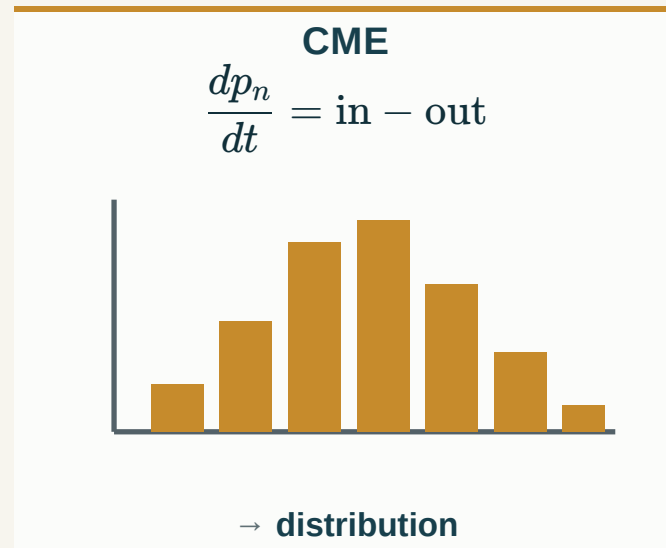
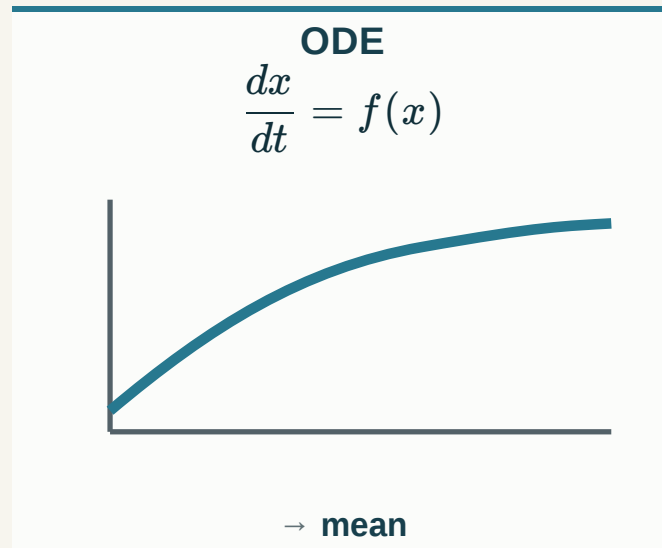
One reaction model, three levels of description: mean, distribution, trajectory.

PRESENTER Yanqing Shi

TEAMMATE Kehan Lang

One reaction model, three views

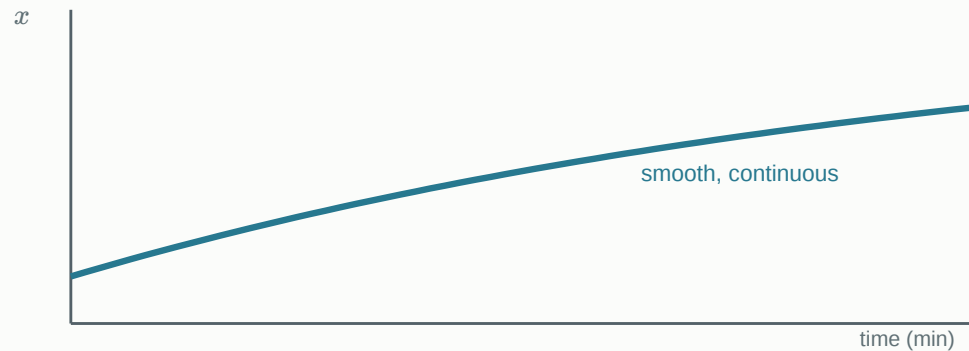
Central idea. The chemical master equation describes a probability distribution; Gillespie's method generates exact sample paths from the same stated stochastic model.



Why is an ODE not enough?

Deterministic rate equation · the mean

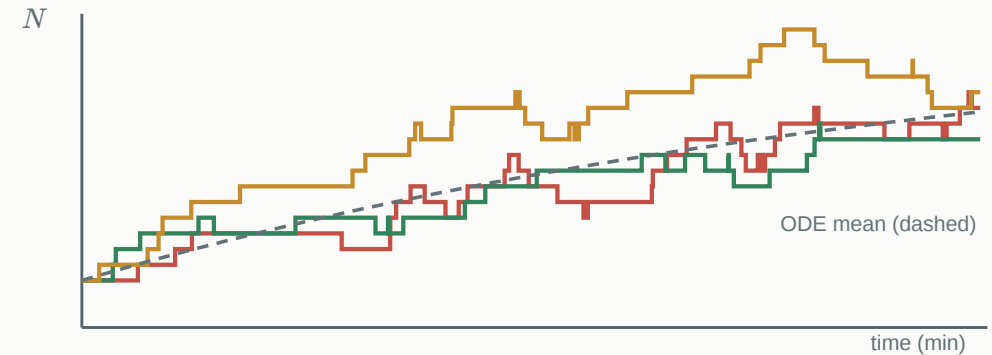
$$\frac{dx}{dt} = b - d x$$



$x(t)$ moves continuously, e.g. from 3 to 3.2 molecules

Single realizations · discrete counts

$$N(t) \in \{0, 1, 2, \dots\}$$

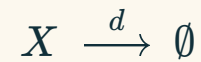
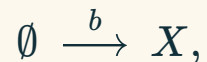


$N(t)$ stays constant, then jumps by ± 1 when a reaction fires

The mean can be smooth, but a single realization jumps. If the state itself is random, track $\Pr\{N(t) = n\}$ instead of only $x(t)$.

Running example

used on every slide from here on



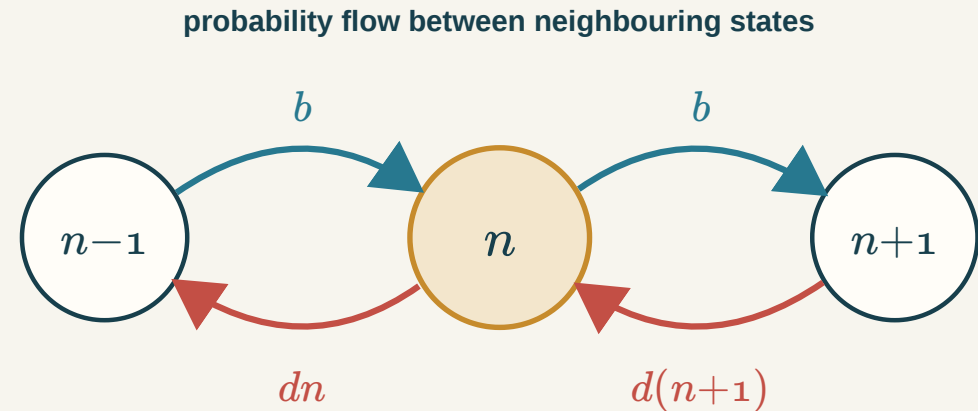
$$b = 4 \text{ min}^{-1}, \quad d = 0.2 \text{ min}^{-1}, \quad N(0) = 3$$

The chemical master equation tracks probability flow

Instead of a single number, the state of knowledge is a probability for every count:

$$p_n(t) = \Pr\{N(t) = n\}$$

Over a short interval dt , probability can **enter** state n from $n - 1$ (a birth) or from $n + 1$ (a death), and **leave** n through a birth or a death. Taking $dt \rightarrow 0$ gives the CME.



blue: birth, constant rate b · **red:** death, rate $d \times$ (number of molecules present)

$$\frac{dp_n}{dt} = \underbrace{b p_{n-1}}_{\text{birth inflow}} + \underbrace{d(n+1) p_{n+1}}_{\text{death inflow}} - \underbrace{(b + d n) p_n}_{\text{outflow}}$$

The same propensities define both the CME and Gillespie

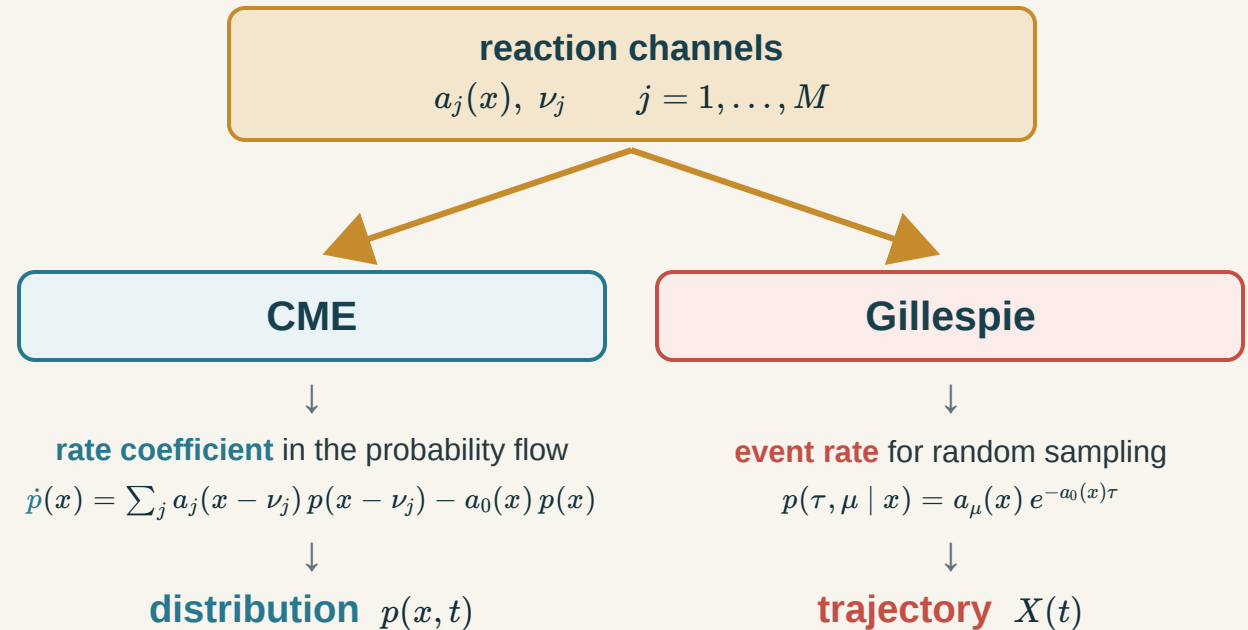
Propensities in the current state $N = n$

$$a_{\text{birth}}(n) = b, \quad a_{\text{death}}(n) = d n$$

$$a_0(n) = b + d n$$

running example, $n = 3$: $a_{\text{birth}} = 4$, $a_{\text{death}} = 0.6$, $a_0 = 4.6 \text{ min}^{-1}$

Meaning. $a_j(x) dt$ is approximately the probability that reaction j fires during the next dt . In general, channel j has a propensity $a_j(x)$ and a state change ν_j ; the CME uses them as flow coefficients, Gillespie as event rates.



CME and Gillespie are not two different models. They use the same reaction channels and the same propensities.

From the CME to the next random event

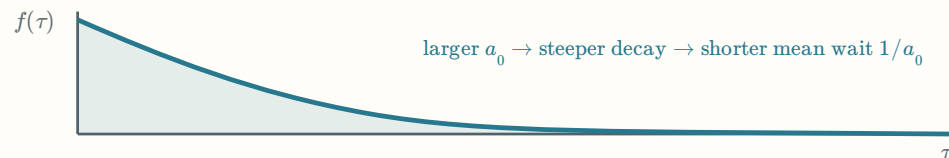
A sample path is fully determined by answering two questions after every event, using the propensities of the current state.

WHEN? time to the next event

$$\tau \sim \text{Exp}(a_0), \quad f(\tau) = a_0 e^{-a_0 \tau}$$

between events the state is fixed, so a_0 is constant and the wait is memoryless;
mean wait $1/a_0$

$$\tau = \frac{1}{a_0} \ln \frac{1}{r_1}$$



$$r_1 \sim U(0, 1)$$

→

$$\tau = \ln(1/r_1)/a_0$$

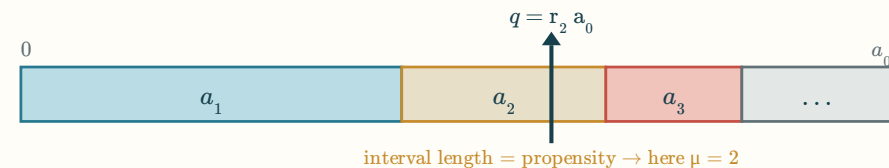
→ **WHEN**

WHICH? identity of the next reaction

$$\Pr(\mu = j) = \frac{a_j}{a_0}$$

relative propensity = selection probability; place $q = r_2 a_0$ on the cumulative-propensity axis

$$\sum_{j=1}^{\mu-1} a_j < r_2 a_0 \leq \sum_{j=1}^{\mu} a_j$$



$$r_2 \sim U(0, 1)$$

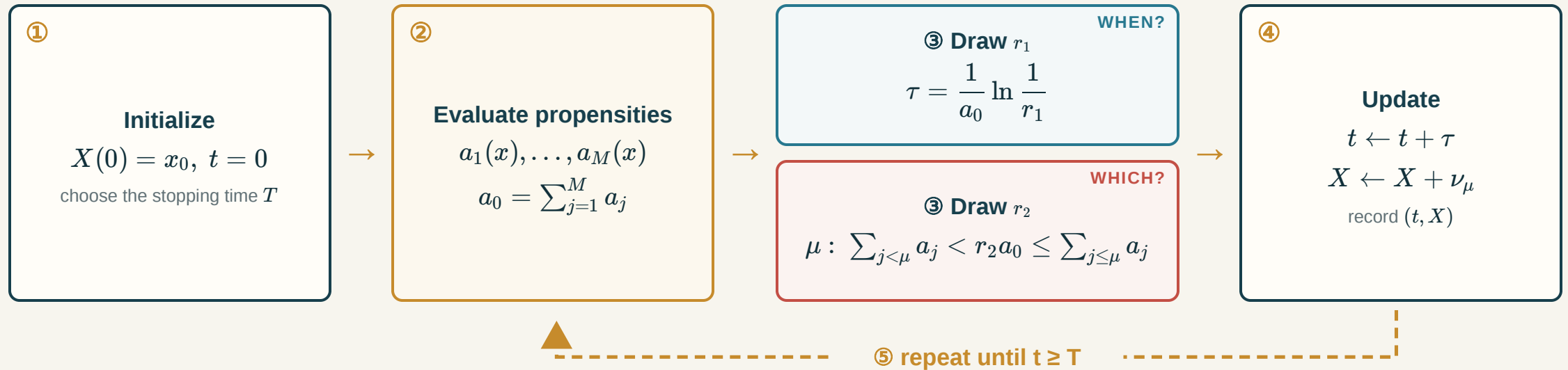
→

$$q = r_2 a_0$$

→

WHICH

Gillespie's Direct Method



Implementation rule. Recalculate every state-dependent propensity after each event. Propensities are constant only during the waiting interval before the next event.

Why “exact”. Steps ③ sample the joint density of the next event implied by the CME, $p(\tau, \mu \mid x) = a_\mu(x) e^{-a_0(x)\tau}$, with no time step and no discretization error.

One stochastic jump, by hand

Current state $n = 3$, rates $b = 4 \text{ min}^{-1}$, $d = 0.2 \text{ min}^{-1}$; the computer draws $r_1 = 0.25$, $r_2 = 0.93$.

1 · propensities

$$a_1 = b = 4$$
$$a_2 = d n = 0.2 \times 3 = 0.6$$
$$a_0 = a_1 + a_2 = 4.6$$

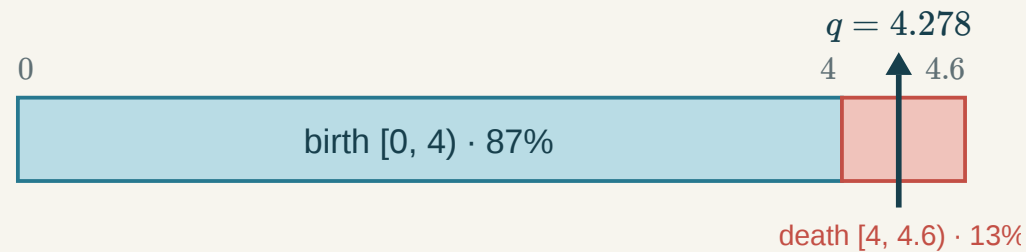
2 · when? use $r_1 = 0.25$

$$\tau = \frac{1}{4.6} \ln \frac{1}{0.25}$$
$$= \frac{1.3863}{4.6} = 0.3014 \text{ min}$$

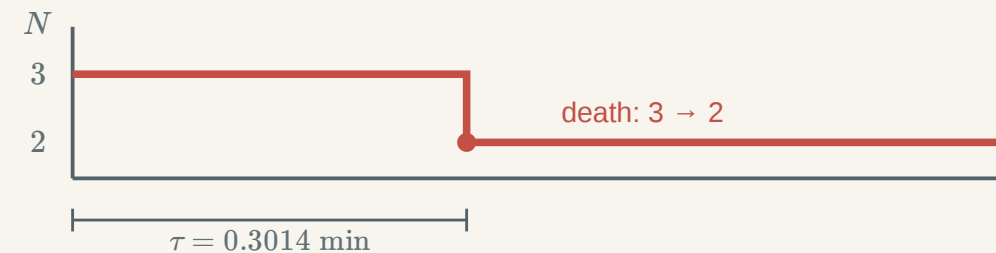
3 · which? use $r_2 = 0.93$

$$q = r_2 a_0 = 0.93 \times 4.6 = 4.278$$
$$q \in [4, 4.6) \Rightarrow \mu = \text{death}$$

which? · cumulative-propensity axis (to scale)



when? · time axis



$t : 0 \rightarrow 0.3014 \text{ min}$

$N : 3 \rightarrow 2$

then recompute: $a_2 = 0.4$, $a_0 = 4.4$, and draw again

ODE, CME, and Gillespie are three levels of description

	ODE	CME	GILLESPIE
State	continuous mean	discrete probability distribution	discrete realization
Output	$x(t)$	$p(x, t)$	$X(t)$
Dynamics	smooth	probability flow	random jumps
Question	What happens on average?	What is the probability?	What does one realization look like?

